

L28: Common Random Numbers and Variance Reduction

Simulation Without a Black Box

Outline

The Comparison Problem

CRN Concept

Variance Reduction Formula

CRN Implementation

Paired-Difference CI and Three-Way Comparison

Key Takeaways

The Comparison Problem: Signal vs. Noise

The Goal

Compare two system designs A and B on the same KPI (e.g., W_q). We want to detect a *real* design difference, not a lucky (or unlucky) random draw.

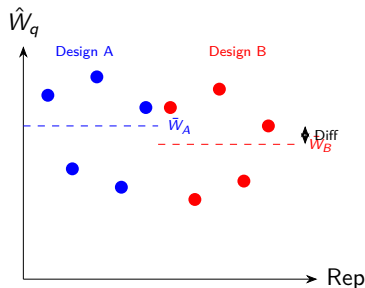
Independent sampling approach:

- Run n replications of A with seeds $s_1, \dots, s_n$
- Run n replications of B with *different* seeds $s_{n+1}, \dots, s_{2n}$
- Estimate $E[W_A - W_B]$ via $\bar{W}_A - \bar{W}_B$

Variance of the difference:

$$\text{Var}(\bar{W}_A - \bar{W}_B) = \frac{\sigma_A^2}{n} + \frac{\sigma_B^2}{n}$$

This can be large when σ_A^2 and σ_B^2 are large — even if $E[W_A - W_B]$ is small.



CRN: Same Random Streams for Both Designs

Common Random Numbers

Use the *same* sequence of random numbers for design A and design B in each replication. Each customer arrives at the same moment and has the same inherent service demand — only the design decision differs.

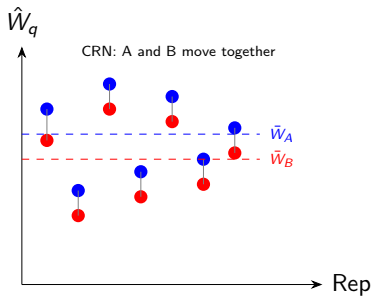
Paired difference estimator:

$$D_i = W_{A,i} - W_{B,i}, \quad i = 1, \dots, n$$

$$E[\bar{D}] = E[W_A] - E[W_B]$$

$$\text{Var}(\bar{D}) = \frac{\sigma_A^2 + \sigma_B^2 - 2 \text{Cov}(W_A, W_B)}{n}$$

If $\text{Cov}(W_A, W_B) > 0$: CRN **reduces variance**.



Variance Reduction: The Math

Variance of $\bar{D}$ Under CRN

$$\text{Var}(\bar{D}) = \frac{\sigma_A^2 + \sigma_B^2 - 2 \text{Cov}(A, B)}{n}$$

Without CRN: $\text{Cov}(A, B) = 0$ (independent runs).

With CRN: $\text{Cov}(A, B) > 0$ if both designs respond similarly to the same random inputs.

Variance reduction ratio (VRR):

$$\text{VRR} = 1 - \frac{\text{Var}_{\text{CRN}}}{\text{Var}_{\text{ind}}} = \frac{2 \text{Cov}(A, B)}{\sigma_A^2 + \sigma_B^2} = 2 \text{Corr}(A, B) \cdot \frac{\sigma_A \sigma_B}{\sigma_A^2 + \sigma_B^2}$$

If $\text{Corr}(A, B) = 0.8$, $\sigma_A \approx \sigma_B$: $\text{VRR} \approx 80\%$ — CRN achieves the same precision with $\sim 5\times$ fewer replications.

When CRN Fails

- Designs respond *oppositely* to randomness: CRN can *increase* variance ($\text{Cov} < 0$)
- Happens when one design benefits from heavy load while the other is hurt by it
- Always check: is $\text{Corr}(A, B) > 0$?

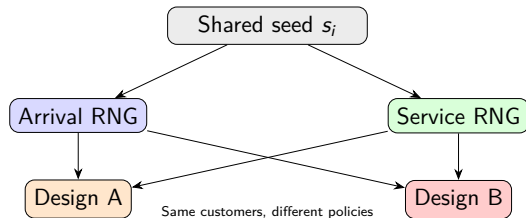
```
corr = np.corrcoef(
    results_A, results_B)[0,1]
print(f"Corr(A,B) = {corr:.3f}")
# If negative: CRN is counterproductive
```

CRN Implementation: Stream Synchronization

Critical requirement: Arrival stream and service stream must be synchronized across designs A and B.

Use separate child generators for each process type so that the k -th customer has the same interarrival and service demand regardless of which design is running.

```
def run_one_crn(seed, design):
    """
    seed: SeedSequence child
    design: "A" or "B"
    """
    # Spawn two sub-streams per rep
    s_arr, s_svc = seed.spawn(2)
    rng_arr = np.random.default_rng(s_arr)
    rng_svc = np.random.default_rng(s_svc)
    # Use rng_arr for interarrivals
    # Use rng_svc for service times
    # Design A/B affects # servers or
    # scheduling rule, not the RNG
    ...
```



Paired-Difference CI and Three-Way Comparison

Paired-Difference 95% CI

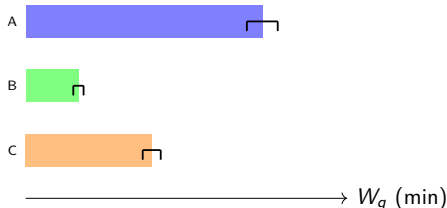
$$\bar{D} \pm t_{n-1, 0.025} \cdot \frac{s_D}{\sqrt{n}}$$

where $D_i = W_{A,i} - W_{B,i}$ and s_D = std of differences.

Three-way comparison example (30 reps, CRN):

Design	$\bar{W}_q$	s	95% CI
A (1 server)	18.4	3.2	[17.2, 19.6]
B (2 servers)	4.1	0.9	[3.7, 4.5]
C (1+trriage)	9.8	1.8	[9.1, 10.5]

With CRN, s_D for B vs. A might be 1.5 (vs. 4.0 without CRN) — the CI is $2.7\times$ narrower.



Warning

With CRN, do *not* use independent-sample t -tests across designs. Use paired t -test on $D_i = W_{A,i} - W_{B,i}$.

Key Takeaways

1. **The comparison problem:** independent runs conflate design differences with random noise.
2. **CRN** uses the same random streams for both designs in each replication, inducing positive correlation.
3. **Variance formula:** $\text{Var}(\bar{D}) = (\sigma_A^2 + \sigma_B^2 - 2 \text{Cov})/n$; CRN reduces variance when $\text{Cov} > 0$.
4. **Implementation:** spawn separate arrival and service sub-streams per replication; share both across designs.
5. **Paired CI:** apply the paired-difference t -CI; do not use the two-sample formula with CRN data.
6. **When CRN fails:** if designs respond oppositely to load, CRN can *increase* variance — always verify $\text{Corr}(A, B) > 0$.

Next Module: M09 — Experimental Design

L29 — 2^k factorial designs for simulation experiments.